Math Related Quiz-Questions

EASY

1. If $f(x)=2x^2-3x+1$, what is $f(2)$?

A) 3
B) 5
C) 7
D) 9

Answer: A) 3

Solution: $f(2)=2(2^2)-3(2)+1=8-6+1=3$.

2. What is the derivative of x^3+4x^2-5x?

A) $3x^2+8x-5$
B) $3x^2+4x-5$
C) x^2+8x-5
D) $3x^2+8x+5$

Answer: A) $3x^2+8x-5$

Solution: Differentiate term by term: $3x^2+8x-5$.

3. What is the determinant of [[2,1],[3,4]]?

A) 5
B) 8
C) 11
D) −5

Answer: A) 5

Solution: $\det=2\times4-1\times3=5$.

4. Evaluate $\lim(x\to2)\ (x^2+3x)$.

A) 10
B) 8
C) 12
D) 6

Answer: A) 10

Solution: Substitute $x=2$: $4+6=10$.

5. If a vector is v=(3,4), what is its magnitude?

A) 5
B) 7
C) 1
D) 25

Answer: A) 5

Solution: $|v|=\sqrt{(3^2+4^2)}=\sqrt{25}=5$.

6. If P(A)=0.4 and P(B)=0.3 for mutually exclusive events, what is P(A∪B)?

A) 0.7
B) 0.12
C) 0.1
D) 1.2

Answer: A) 0.7

Solution: $P(A\cup B)=0.4+0.3=0.7$.

Medium

1. Solve $x^2-5x+6=0$.

A) x=2,3
B) x=−2,−3
C) x=1,6
D) x=−2,3

Answer: A) x=2,3

Solution: $(x-2)(x-3)=0$, so x=2,3.

Medium 2. If $y=x^2e^x$, find dy/dx.

A) $e^x(x^2+2x)$
B) $2xe^x$
C) x^2e^x
D) $e^x(x^2-2x)$

Answer: A) $e^x(x^2+2x)$

Solution: Product rule: $y'=2xe^x+x^2e^x=e^x(x^2+2x)$.

3. Find the eigenvalues of [[4,0],[0,−2]].

A) 4 and −2

B) 2 and −4
C) 4 and 2
D) −4 and −2

Answer: A) 4 and −2

Solution: For a diagonal matrix, the diagonal entries are the eigenvalues.

4. Two fair dice are rolled. What is the probability that their sum is 9?

A) 1/9
B) 1/6
C) 1/12
D) 1/18

Answer: A) 1/9

Solution: Four favorable outcomes out of 36: 4/36=1/9.

5. Solve 2x+y=7 and x−y=2.

A) x=3,y=1
B) x=2,y=3
C) x=1,y=5
D) x=4,y=−1

Answer: A) x=3,y=1

Solution: Adding gives 3x=9, so x=3; then y=1.

Hard

1. Find lim(x→0) (sin 3x)/x.

A) 3
B) 1
C) 0
D) 1/3

Answer: A) 3

Solution: 3·sin(3x)/(3x)→3.

2. If A=[[1,2],[3,4]], what is det(A^2)?

A) 4
B) −4
C) −2
D) 16

Answer: A) 4

Solution: $\det(A)=-2$, so $\det(A^2)=(-2)^2=4$.

3. Find the critical points of $f(x)=x^3-3x^2-9x+5$.

A) $x=-1,3$
B) $x=1,-3$
C) $x=-3,1$
D) $x=0,3$

Answer: A) $x=-1,3$

Solution: $f'=3(x-3)(x+1)$; hence $x=3,-1$.

4. Evaluate $\int_0^1 x \ln(x)\, dx$.

A) $-1/4$
B) $1/4$
C) $-1/2$
D) 0

Answer: A) $-1/4$

Solution: Integration by parts gives $(x^2/2)\ln x - x^2/4$; the definite value is $-1/4$.